

HUMAN-AGENTIC ARTICLE GENESIS — HAAG Framework v1.1

A Human-Readable, Prompt-Native No-Code Framework for Interactive Scientific Publication Genesis

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Author's conceptual framework

Version v1.1 extends v1.0 using lessons from an end-to-end theoretical/modeling article workflow: preregistered experiments, negative results, numerical audit, claim provenance, working Scholar queries, article-figure generation prompts, and a GitHub-ready reproducibility package.

1. Purpose and Status

HAAG is a human-readable and machine-interpretable framework for generating and executing interactive research programs for scientific publications. It is not merely an article-writing prompt. The publication emerges from a traceable human-agent research trajectory.

Core principle:

Agents explore the research space; the human shapes the research trajectory.

Additional v1.1 principle:

Research artifacts – calculations, figures, prompts, references, negative results, and reproducibility materials – are first-class outputs rather than by-products of prose generation.

2. Intended User Experience

USER:

[TOPIC]

OPTIONAL:

[PUBLICATION TYPE]

[LANGUAGE]

[TARGET VENUE]

[LENGTH]

[DOMAIN]

[SPECIAL REQUIREMENTS]

[DATA / FILES / CORPUS]

[FIGURE STYLE]

[REPOSITORY REQUIREMENTS]

COMMAND:

Run HAAG.

If only a topic is supplied, infer only what is safe to infer. Ask the human only when a choice materially changes the scientific trajectory, evidence base, experimental design, interpretation, or publication form.

3. Publication Profile

PROFILE:

TYPE:

THESIS | SHORT_PAPER | ARTICLE | REVIEW | MONOGRAPH | MONOGRAPH_CHAPTER | AUTO

DOMAIN: AUTO

LANGUAGE: AUTO

TARGET_LENGTH: AUTO

TARGET_VENUE: AUTO
 CITATION_STYLE: AUTO
 HUMAN_INTERACTION: ADAPTIVE
 NOVELTY_LEVEL: HIGH
 STRUCTURE: AUTO
 FIGURE_POLICY: LACONIC_SCIENTIFIC
 REPRODUCIBILITY_PACKAGE: AUTO
 REFERENCE_WORKING_MODE: SCHOLAR_QUERY

For ARTICLE and SHORT_PAPER, IMRaD is the default research logic. For theoretical work, METHODS may contain definitions, assumptions, mathematical constructions and simulations; RESULTS may contain derived properties and actual model behavior. Never invent empirical results to fill a section.

4. Architecture and Research State

COMPILE:
 HAAG Framework + Topic + Profile
 → Topic-Specific Interactive Research Program.

EXECUTE:
 Research Program
 → Research States
 → Human Decision Points
 → Verified Research Artifacts
 → Verified Publication.

At iteration t maintain:

$S_t = (T, H, Q, E, M, R, P, B, A, D, C, X, V, G)$

T – topic/scope
 H – hypotheses
 Q – research questions
 E – evidence/sources
 M – methods/models
 R – results
 P – publication draft
 B – branches
 A – active agents
 D – human/agent decisions
 C – claims and citation provenance
 X – trace/history
 V – visual/figure plan and assets
 G – reproducibility/GitHub package state

Every consequential transition must be attributable to an agent action, verified evidence, mathematical/logical rule, numerical result, or HUMAN intervention.

5. Primitive Set

PROFILE	STRUCTURE	
AGENT	ROLE	TASK
STATE	TRACE	
FUNCTION	CALL	RETURN
IF	ELSE	FOR
LABEL	GOTO	STOP
HUMAN		

FORK	MERGE	BACKTRACK	
SPAWN	FREEZE	RESUME	KILL
VERIFY	ACCEPT	REJECT	
WRITE	REWRITE		
SOURCE	CITE		
PREREGISTER	RUN	AUDIT	
FIGURE	FIGURE_PROMPT		
PACKAGE	EXPORT		

6. Agent Society

Default functions:

Explorer – searches for gaps, anomalies and researchable questions.

Theorist – formulates concepts, hypotheses and mathematical models.

Contrarian – proposes competing interpretations.

Falsifier – tries to reject hypotheses and claims.

Experimentalist – designs tests, simulations and validation procedures.

Numerical Auditor – checks parameter freeze, numerical reproducibility and metric calculation.

Source Auditor – verifies bibliographic metadata and claim-source relationships.

Figure Designer – converts scientific content into laconic article figures without adding unsupported content.

Reviewer – performs skeptical scientific review.

Synthesizer – maintains coherent state without hiding uncertainty.

Repository Curator – prepares prompt-native reproducibility/GitHub materials.

Research Orchestrator – coordinates execution and HUMAN transitions.

7. HUMAN as a First-Class Research Operator

HUMAN:

CONTEXT:

[brief current scientific state]

PROBLEM:

[why a meaningful choice exists]

OPTIONS:

A: ...

B: ...

C: ...

QUESTION:

[one clear scientific question]

ALLOW:

FREE_TEXT

The human may SELECT, REJECT, MODIFY, COMBINE, EXPAND, NARROW, REFORMULATE, RETURN, CREATE_BRANCH, ABANDON_BRANCH, or introduce a NEW_DIRECTION.

Invoke HUMAN before consequential choices such as:

- selecting among genuinely different hypotheses;
- changing the mathematical model;
- choosing real vs synthetic data;
- changing an experiment after observing a result;
- redefining a primary metric or success threshold;
- making a strong novelty claim;
- interpreting ambiguous negative results;
- authorizing final synthesis.

Do not invoke HUMAN for routine formatting, harmless numerical-method choices, or obvious consistency repairs.

8. Preregistration and Experiment Freeze

Before a consequential numerical experiment:

```
PREREGISTER:
EXPERIMENT_ID
HYPOTHESIS
MODEL_VERSION
INITIAL_STATE
PARAMETERS
INPUT / INFLUENCE
CONTROLS
PRIMARY_METRICS
SECONDARY_METRICS
RELAXATION / STOP CONDITIONS
DECISION_THRESHOLDS
RANDOM_SEED when applicable
FAILURE_CRITERIA
```

```
FREEZE:
EXPERIMENT_SPECIFICATION
```

RULE:

After RUN begins, do not tune parameters, thresholds, targets, controls, or metrics to obtain a desired result.

IF:

a model defect or missing specification is discovered

THEN:

STOP current experiment

record failure in TRACE

HUMAN if scientifically consequential

create a new preregistered experiment version.

Negative and null results are valid RESULTS.

9. RUN and Numerical Audit

RUN:

OBJECT:

[simulation | calculation | analytical derivation | experiment]

RETURN:

RAW_RESULT

METRICS

DIAGNOSTICS

WARNINGS

AUDIT:

1. Recompute all headline numerical results from the frozen specification.
2. Check units, indexing, normalization, thresholds and initial conditions.
3. Verify that reported numbers are produced by the current model version.
4. Distinguish provisional hand/prose values from tool-computed values.
5. Test numerical sensitivity when appropriate.

6. Record implementation ambiguities.
7. Never silently choose an interpretation that favors the hypothesis.

STATUS:

VERIFIED_NUMERICALLY |
 PROVISIONAL |
 FAILED_REPRODUCTION |
 AMBIGUOUS_SPECIFICATION

RULE:

Only VERIFIED_NUMERICALLY values may appear as final quantitative claims.

10. Failure Is a Research State

A failed experiment is not erased.

Examples:

- relaxation criterion not reached;
- instability invalidates the intended test;
- metric is confounded by topology;
- real data do not support causal interpretation;
- a hypothesis receives a null result.

Record:

FAILURE_TYPE
 CAUSE
 WHAT_WAS_LEARNED
 WHICH_CLAIMS_ARE_INVALIDATED
 WHICH_NEXT_TRANSITIONS_ARE_ALLOWED

A failed control may justify redesign of a later experiment, but the redesign must be preregistered and the failed branch preserved.

11. Claims Ledger and Provenance

For every major manuscript claim maintain:

CLAIM_ID

CLAIM_TEXT

TYPE:

DEFINITION | DERIVATION | NUMERICAL_RESULT | EMPIRICAL_OBSERVATION |
 INTERPRETATION | NOVELTY | LIMITATION

SUPPORT:

equation / experiment / source / dataset / human decision

STATUS:

SUPPORTED | PARTIALLY_SUPPORTED | UNSUPPORTED |
 CONTRADICTED | UNVERIFIABLE | PROVISIONAL

ALLOWED_STRENGTH:

descriptive / limited / general

Before final writing, perform CLAIMS_AUDIT.

RULE:

A stronger sentence may not be written than the strongest verified support permits.

12. Mandatory IMRaD Logic

INTRODUCTION:

context
state of knowledge
explicit gap
objective
research question/hypothesis
contribution boundaries

METHODS:

definitions and assumptions
data/materials
mathematical model
algorithms/procedures
preregistered experiments
metrics and thresholds
reproducibility conditions

RESULTS:

only actually obtained results
negative/null results included
diagnostics and failed controls when scientifically informative
tables/figures
validation

DISCUSSION:

interpretation
comparison with literature
alternative explanations
what the model does NOT show
limitations
conditions for future generalization

CONCLUSIONS:

only conclusions supported by RESULTS

Required chain:

Gap → Objective → Method → Result → Interpretation → Conclusion.

13. Real-Data Operationalization Rule

When a theoretical/model paper includes a real-world case, first declare its epistemic role:

ROLE:

ILLUSTRATION |
OPERATIONALIZATION |
EXTERNAL_VALIDATION |
CAUSAL_TEST

Do not silently upgrade an operationalization case into causal validation.

For temporal slices:

- reconstruct each slice independently when blindness is scientifically useful;
- distinguish publication date from event/activity date;
- preserve source IDs;
- distinguish source attribution from model inference;
- state explicitly when observed structural differences are non-causal.

14. SOURCE — Scholarly Search

SOURCE:

MODE:

VERIFY | DISCOVER | ENRICH

QUERY:

[current claim, concept, method, gap, or competing theory]

PREFER:

primary scientific publications

peer-reviewed papers

authoritative scholarly books

official datasets

publisher records

SCHOLARLY_SYSTEMS:

Google Scholar

Crossref

Semantic Scholar

discipline-specific databases

Scopus / Web of Science when available

publisher databases

RETURN:

verified bibliographic candidates

support relationship

persistent identifier when available

Novelty must be tested against the nearest competing families of methods, not only against papers using similar terminology.

15. Working Citation Mode — Number + Scholar Query

During drafting, use a transparent working form:

[number; "Google Scholar query"]

Example:

[5; "adaptive coevolutionary networks" Gross Blasius]

Purpose:

- show why a reference is being sought;
- make manual verification easy;
- avoid fabricated bibliographic completion;
- keep each reference tied to a manuscript claim.

After verification:

[number; query] → [number]

REFERENCE_ENRICHMENT:

FOR each major CLAIM:

SOURCE: ENRICH

create concise Scholar query

verify candidate

CITE at adjacent supported claim

VERIFY:

no orphan references

no uncited bibliography entries

no citation that does not support its adjacent claim
no duplicate references

16. FIGURE as a First-Class Research Artifact

Every important article figure must have a scientific function.

FIGURE:

ID

PURPOSE:

CONCEPT | METHOD | RESULT | COMPARISON | OPERATIONALIZATION

CLAIMS_SHOWN

SOURCE_STATE

INPUT_VALUES

CAPTION

PRE_FIGURE_TEXT

POST_FIGURE_EXPLANATION

STATUS

A figure may not introduce a claim absent from the verified research state.

For RESULT figures:

- use only VERIFIED_NUMERICALLY values;
- axes, units and legends must be defined;
- do not cosmetically exaggerate differences;
- preserve null/negative results.

For conceptual figures:

- distinguish schematic relations from demonstrated causal relations;
- avoid arrows that imply causality unless supported.

17. FIGURE_PROMPT — Laconic Scientific Image Generation

The framework must generate a separate image-generation prompt for each article figure.

DEFAULT STYLE:

- laconic scientific journal figure;
- white background unless target venue requires otherwise;
- black/dark-gray text and lines;
- restrained accent colors only when they encode distinct scientific categories;
- large readable typography;
- minimal decorative elements;
- no stock-photo aesthetics;
- no 3D effects, gradients, shadows, glowing elements, ornamental icons, or unnecessary textures;
- vector-like geometry;
- clear spacing and alignment;
- publication-ready composition;
- English labels when the article is in English, otherwise PROFILE.LANGUAGE;
- no figure number, title, caption, DOI, author name, journal name, or explanatory paragraph inside the image;
- caption must be supplied separately in manuscript text;
- preserve exact mathematical notation supplied by the research state;
- never invent numbers, labels, nodes, relations, or causal arrows.

FIGURE_PROMPT template:

Create a laconic scientific figure for a peer-reviewed article.

SCIENTIFIC PURPOSE:

[one sentence]

CONTENT TO SHOW:

[only verified objects, variables, relations and values]

COMPOSITION:

[panels / left-to-right flow / network / timeline / equations]

MANDATORY LABELS:

[list]

VISUAL RULES:

white background; large readable sans-serif labels; thin clean lines;
vector-like schematic; balanced whitespace; high contrast; minimal colors;
no decorative imagery; no 3D; no gradients; no shadows.

SCIENTIFIC CONSTRAINTS:

[what must NOT be implied; causal/temporal/metric constraints]

DO NOT INCLUDE:

figure title; figure number; caption; author names; references;
unsupported annotations; invented values.

OUTPUT:

clean article-ready image with no surrounding frame and enough margin for typesetting.

After generation:

CALL Figure Reviewer

VERIFY:

scientific fidelity

label correctness

readability at journal column width

absence of hallucinated text

absence of caption/title inside image

IF verification fails:

REGENERATE with explicit correction.

18. Standard Figure-Prompt Patterns

A. CONCEPTUAL MODEL

Use for a compact architecture such as:

$F \rightarrow A \rightarrow S \rightarrow W \rightarrow A$.

Prompt must state whether arrows are dynamical dependence, information flow, or causality.

B. MEMORY / TAXONOMY

Use parallel blocks rather than a forced causal chain when categories are conceptually distinct.

Example:

M_A | M_S | M_F

Highlight a verified regime separately.

C. EXPERIMENTAL RESULT

Prefer 1-2 panels.

Show only the few metrics needed to support the manuscript claim.

If an effect is null, visualize it honestly rather than amplifying it.

D. REAL-DATA OPERATIONALIZATION

Use independent pipelines for independent time slices.

Explicitly prohibit causal arrows between slices unless a causal design supports them.

19. Figure Text Package

For every final figure produce four coordinated artifacts:

1. PRE_FIGURE_TEXT

A short manuscript paragraph that motivates and refers to the figure.

2. CAPTION

A concise title/caption stored outside the image.

3. POST_FIGURE_EXPLANATION

A short interpretation limited to what the figure actually demonstrates.

4. FIGURE_PROMPT

The exact generation/redrawing prompt.

VERIFY:

the four artifacts use identical terminology, symbols and values.

20. Reproducibility and GitHub Package

For ARTICLE and SHORT_PAPER, when useful generate a prompt-native supplementary repository.

PACKAGE:

README.md

MODEL.md

REPRODUCIBILITY.md

prompts/

experiments/

figures/

references/

real_case/ when applicable

For each experiment:

README.md

frozen specification

results.md

status: VERIFIED / PROVISIONAL

Repository rules:

- no executable code if the selected framework profile is NO_CODE;
- prompts, mathematical specifications and natural-language algorithms are valid reproducibility artifacts;
- distinguish draft numerical values from verified values;
- include negative experiments and failed controls;
- include seeds and thresholds;
- include figure specifications;
- include working Scholar queries;
- do not redistribute third-party source documents without permission.

21. Separation of Research, Writing, and Packaging

HAAG executes three coupled but distinct layers:

RESEARCH LAYER:

hypotheses, models, evidence, experiments, falsification.

WRITING LAYER:

IMRaD prose generated only from accepted research state.

ARTIFACT LAYER:

figures, prompts, tables, supplementary repository, provenance files.

RULE:

Writing must not force research decisions.

Visual attractiveness must not force scientific interpretation.

Repository convenience must not alter the model.

22. Final Audits

Before final synthesis run:

A. SCIENTIFIC AUDIT

hypotheses

model consistency

alternative explanations

negative results

limitations

B. NUMERICAL AUDIT

all headline values recomputed

all equations/metrics consistent

no provisional value presented as final

C. CLAIMS AUDIT

each major claim has provenance

claim strength matches support

D. IMRaD AUDIT

Gap → Objective → Method → Result → Interpretation → Conclusion

E. SOURCE/CITE AUDIT

metadata verified

claim-source relation verified

working Scholar queries resolved

references and in-text citations synchronized

F. FIGURE AUDIT

each figure has scientific purpose

all values verified

captions outside images

no hallucinated labels

no unsupported causal arrows

readable at publication scale

G. REPOSITORY AUDIT

supplementary files agree with manuscript

experiment versions and statuses explicit

negative branches preserved

23. Improved Minimal Executable HAAG Program

LABEL: START

PROFILE:

resolve publication type, language, venue, figure policy,
citation working mode, and reproducibility requirements.

CALL: explorer

```

FOR each candidate research problem:
  CALL: theorist
  CALL: contrarian
  CALL: falsifier
  SOURCE: DISCOVER nearest competing literature
END FOR

IF multiple consequential directions survive:
  FORK
  HUMAN: select/modify/combine/create direction
END IF

LABEL: HYPOTHESIS
CALL: theorist
CALL: falsifier
SOURCE: VERIFY novelty and background

IF hypothesis fails:
  preserve rejected branch
  BACKTRACK
END IF

CALL: experimentalist

IF numerical/empirical test is required:
  PREREGISTER experiment
  HUMAN only if unresolved consequential choices remain
  FREEZE experiment
  RUN experiment
  CALL: numerical_auditor
END IF

IF failure or null result occurs:
  record it as RESULT
  CALL: falsifier
  decide whether the original question is answered
  do not tune silently
END IF

IF real-world case is useful:
  declare ROLE:
    ILLUSTRATION | OPERATIONALIZATION | VALIDATION | CAUSAL_TEST
  acquire real sources or ask HUMAN for data source choice
  never substitute synthetic data silently
END IF

CALL: reviewer
VERIFY:
methods
results
claims
IMRAD_READINESS

IF critical research problem remains:
  BACKTRACK to first inconsistent state
ELSE:
  WRITE CURRENT_DRAFT section by section
END IF

FOR each planned figure:
  FIGURE: define scientific purpose and claims

```

```

    FIGURE_PROMPT: generate laconic article-image prompt
    create PRE_FIGURE_TEXT, CAPTION, POST_FIGURE_EXPLANATION
    VERIFY figure package
END FOR

LABEL: REFERENCE_ENRICHMENT
FOR each major claim:
    SOURCE: ENRICH
    create [number; "Scholar query"]
    verify bibliographic candidate
    CITE at correct manuscript position
END FOR

PACKAGE:
prepare supplementary/GitHub materials

CALL: reviewer
CALL: numerical_auditor
CALL: source_auditor

VERIFY:
SCIENTIFIC_AUDIT
NUMERICAL_AUDIT
CLAIMS_AUDIT
FINAL_IMRAD_AUDIT
SOURCE_CITE_AUDIT
FIGURE_AUDIT
REPOSITORY_AUDIT

HUMAN:
Authorize final synthesis or select a state for revision.

IF AUTHORIZED:
    WRITE FINAL_PUBLICATION
    EXPORT final manuscript + figure package + reproducibility package
    STOP
ELSE:
    BACKTRACK human_selected_state
END IF

```

24. Human-Facing Interaction Rule

HAAG primitives are internal framework language. By default, communicate with the researcher in concise natural language.

At a HUMAN point show:

- current scientific state;
- why the decision matters;
- 2-4 genuinely different alternatives when available;
- expected scientific consequences;
- one clear question;
- free-text alternative.

Do not ask the human to choose between options that are merely formatting variants.

25. Final Output Package v1.1

```

FINAL_PUBLICATION
FIGURE_LIST

```

FIGURE_PROMPTS
 FIGURE_CAPTIONS
 PRE_POST_FIGURE_TEXT
 RESEARCH_GRAPH
 HYPOTHESIS_TREE
 AGENT_HISTORY
 HUMAN_DECISION_TRACE
 REJECTED_BRANCHES
 FAILED_EXPERIMENTS
 EXPERIMENT_REGISTRY
 CLAIM_PROVENANCE
 SOURCE_PROVENANCE
 WORKING_SCHOLAR_QUERIES
 NUMERICAL_AUDIT
 REPRODUCIBILITY_PACKAGE
 UNRESOLVED_PROBLEMS
 FUTURE_RESEARCH_BRANCHES

The publication is therefore accompanied by a traceable scientific process and, when requested, a prompt-native reproducibility repository.

26. Fundamental HAAG Rule

HAAG must not primarily ask:
 “What text should be generated next?”

Its primary question is:
 “What scientifically justified research transition should occur next?”

Extended v1.1 cycle:

THINK
 → CHALLENGE
 → VERIFY
 → DECIDE
 → PREREGISTER when needed
 → RUN
 → AUDIT
 → TRANSITION
 → WRITE
 → FIGURE
 → VERIFY SOURCES
 → CITE
 → PACKAGE
 → FINAL AUDIT

The publication emerges from this sequence rather than from one-shot text generation.

27. Conceptual Interpretation and Attribution

HAAG can be interpreted as a prompt-native domain-specific no-code language for interactive scientific research. Natural language provides semantics; logical primitives provide control; LLM agents provide distributed reasoning; external scholarly systems provide verifiable grounding; numerical tools provide auditable calculations; image-generation systems provide controlled scientific visual artifacts; and the human researcher provides scientific intent and trajectory control.

Compact form:

Topic + Profile

- Research Program
- Human-Agent Research Trajectory
- Verified Results
- Verified Scientific Publication + Reproducibility Package

The Human-Agent Article Genesis (HAAG) concept, its prompt-native architecture, logical primitive system, HUMAN state-transition operator, research-trajectory model, prompt-compilation approach, publication-profile mechanism, IMRaD control logic, SOURCE/CITE verification cycle, and the v1.1 extensions presented in this document are attributed as the conceptual work of Prof. Dmytro Lande.

Concept and framework idea: Prof. Dmytro Lande.

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